Transparency without pricing: a credit-level forensic account of collapse in the first tokenized carbon pool

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arbon credits are supposed to pay for real climate benefits, but their quality varies enormously. Economic theory warns that when goods of mixed quality sell at one price, the bad drive out the good. This is the classic lemons problem, and the standard cure is transparency: let buyers see quality, and the market should protect itself. We test that cure in the first market that genuinely tried it. The Base Carbon Tonne pool placed 22 million tonnes of carbon credits on a public blockchain, where anyone could inspect every credit, and sold them all at a single price. The pool collapsed anyway. To find out why, we read its complete transaction record: every deposit, every withdrawal, every account, over its whole life. The collapse was widely blamed on low-quality forest-conservation (REDD+) credits, but those made up only 4% of the pool. What filled it instead, 69%, were old renewable-energy credits that bring little additional climate benefit. A few accounts then extracted most of the valuable credits, leaving 9.3 million tonnes of low-value ones stranded to date. Credits that also sat in a sister pool, one that screened deposits for quality, fared very differently there, which points to pool design, not credit type, as what decided their fate. Transparency alone, we conclude, appears insufficient. Markets that pool assets of mixed quality need to price quality, not merely disclose it. That warning applies to carbon policy under the Paris Agreement and to the 25-billion-dollar industry now tokenizing other real-world assets on the same single-price design.

Introduction

Companies and governments buy billions of dollars of carbon credits each year to offset their emissions. Each credit is supposed to stand for one tonne of CO_2 kept out of the atmosphere. In reality, the climate value behind that promise varies enormously from credit to credit. The central question for any carbon market is whether it can tell the difference.

When a market cannot price such differences, economics has a name for what follows. Adverse selection, formalized by Akerlof in 1970 as the "lemons problem", is the dynamic in which low-quality goods crowd out high-quality ones[1,2,3]. The voluntary carbon market (VCM) faces exactly this risk. Its annual transaction value peaked above $2 billion in 2021–2022, yet fewer than 16% of credits from investigated projects represent genuine reductions[4], and at least 52% of Indian wind power offsets went to projects that would have been built anyway[5].

Blockchain-based tokenization promised a way out: radical transparency. Tokenized carbon pools record every transaction on a public ledger, so any participant can inspect a credit's quality before trading. The first and largest such experiment was the Base Carbon Tonne (BCT) pool, run by Toucan Protocol. It accepted credits verified under the Verra registry, the world's largest carbon-credit standard, and turned each one into an interchangeable token. Every token sold at the same price, whatever the quality behind it. The pool token's price fell from a late-2021 peak above $8 per tonne to $0.71 by mid-2023, and to effectively zero by 2026.

Commentators and academics blamed the collapse on low-quality forest conservation credits, specifically REDD+, a program that pays landowners to avoid deforestation[2,6,7,8,9]. The story fit a literature full of REDD+ crediting problems[4,10,11]. But nobody checked it against the pool's actual contents, which are publicly recorded and straightforward to query. A deeper gap sits behind that omission: the assumption that transparency prevents quality collapse has never been empirically tested anywhere, because conventional carbon markets trade over the counter and leave no public record to test against.

Here we test it. We analyze every deposit (1,187), every recorded pool withdrawal within the scored tokens (35,432), every credit token (161 of 345 individually scored on our six-dimension framework; Methods), and every market participant (509 depositors, 28,897 redeemers) across BCT's operating history. The data contradict the prevailing narrative. BCT was not full of forest credits; the blamed REDD+ type was just 4.2% of it. It was 69.1% renewable energy credits, mostly grid-connected hydropower from China, India, Turkey, and Brazil, with near-zero additionality, meaning the projects would have been built with or without carbon revenue. A few accounts then extracted most of the high-demand credits (1.55 million tonnes), while 9.3 million tonnes of low-quality credits remain stranded to date.

A second comparison sharpens the finding. Some credits sat in two pools at once, one screened for quality and one not. The same credits were 100% redeemed from unscreened BCT but only 28.5% from the quality-screened pool, a contrast consistent with pool design, not credit type, shaping asset fate. Unlike prior work, which quantifies overcrediting at the project level[4,5,22,23], our analysis traces how overcredited assets are selectively extracted inside a pooled market even when quality is publicly observable.

We call this outcome *design-enabled adverse selection*. It differs from Akerlof's classical lemons problem, which requires private information about quality. Here everyone could see the quality; prices simply could not respond to it. The pool treats every deposited credit as identical: a participant cannot buy a specific high-quality credit, only a generic pool unit priced at the pool average, though a redeemer may choose which underlying credit that unit withdraws. We argue that uniform pricing of heterogeneous assets under voluntary participation is sufficient to produce market failure.

Because every transaction is publicly recorded, the complete ledger supports the first credit-level forensic account of this dynamic, a reconstruction that opaque off-chain markets cannot supply. The implication reaches beyond carbon: it covers the $25-billion-plus sector now tokenizing other real-world assets, and Article 6.4, the UN crediting mechanism under the Paris Agreement. Transparency without price differentiation appears insufficient for market function.

Results

BCT's real composition contradicts the REDD+ narrative

We extracted all 1,187 deposit records from BCT's public transaction ledger, covering 168 Verra VCS (Verified Carbon Standard) projects and about 22 million tonnes (Methods).

Renewable energy credits dominated the pool (Fig. 1). Grid-connected hydro, wind, and solar projects (predominantly from China and India, with Turkey and Brazil) registered between 2008 and 2013 constituted 69.1% of total deposited tonnage. These are CDM-era categories (the Clean Development Mechanism, the Kyoto Protocol's now-wound-down UN crediting program, 2004–2020) that the ICVCM (Integrity Council for the Voluntary Carbon Market, the sector's main governance body) has declined to approve for its Core Carbon Principles (CCP) label. Multiple independent studies document their near-zero additionality: Cames et al.[12] and Schneider[13] in program-level assessments, Probst et al.[4] in a meta-analysis finding no statistically significant emission reductions from wind power CDM projects, and Calel et al.[5] estimating 52% non-additionality among Indian wind offsets. The projects would have been built regardless of carbon revenue.

REDD+ credits, widely blamed for BCT's collapse, constituted just 4.2% of the pool, about one-sixteenth of the renewable share. Even with all nature-based categories combined (REDD+, ARR (afforestation, reforestation, and revegetation), and IFM (improved forest management)), the total reaches only 10%, versus 69.1% for renewables alone.

Base-rate selection: BCT over-selected renewables at 1.87× the registry rate

BCT's renewable dominance could reflect the broader carbon credit supply rather than a selection effect. We obtained Verra VCS issuance data from MSCI[21] and Ecosystem Marketplace[24] reports indicating that renewables constitute approximately 37% of total VCS issuance by volume (central estimate; sensitivity range 26–48%).

BCT's renewable share (69.1% by tonnage) is 1.87× the VCS base rate (a ratio we compute; 1 = no over-selection). Because deposits cluster by account, we use account-clustered inference rather than a naive binomial test: an account-level permutation test (10,000 iterations) produces $p < 0.0001$, and 89.0% of accounts hold majority-renewable portfolios. The over-selection is robust to conservative base-rate assumptions (remaining 1.44× at a 48% null) and to account-level bootstrap and concentration-adjusted (HHI) tests (Supplementary Methods).

Conversely, REDD+ is under-represented at 0.2× the VCS base rate (BCT 4.2% vs. VCS ∼17%), directly contradicting the narrative that REDD+ credits overwhelmed the pool.

Bridge-level decomposition. Toucan's bridge, the mechanism that moves an off-chain credit on-chain, passed essentially every Verra-bridged credit into BCT: 93.5% of tokens and 99.6% of tonnage (Supplementary Methods). The 1.87× over-selection is therefore a bridge-level phenomenon. The permissionless bridge, open to anyone, disproportionately attracted renewable credits from Verra; the pool itself had essentially no selection space.

Price-quality feedback loop: composition predicts price collapse

We next ask whether quality degradation drove the price collapse. We merged daily BCT prices in US-dollar terms (1,493 daily observations, October 2021 to July 2026; Methods) with daily cumulative quality metrics from the deposit stream ($n = 331$ overlapping days).

Correlation. BCT's price falls as pool quality falls. Price and the cumulative tonnage-weighted mean quality score are strongly correlated (Pearson $r = 0.77$, $p < 10^{-66}$; Fig. 2a). For pool-level indices we report the score's inverse, the Pool Quality Deficit (PQD = 1 − score/100), against which the same correlation is −0.77 (Methods). In a first-differenced OLS regression (analyzing day-to-day changes rather than levels, the credible specification for non-stationary series), each percentage-point increase in the renewable share of deposits predicts a \$0.018 decrease in BCT's price ($\beta = -1.8$ on the 0–1 share, $p < 0.001$). Exploratory weekly Granger (temporal-precedence) tests are asymmetric, with the dominant channel running from price to quality (Supplementary Methods).

Redemption-side evidence. The 35,432 outbound transfer (redemption) events show that selection operated on both sides. Non-renewable credits were preferentially redeemed: industrial gas 100%, REDD+ 99.8%, IFM 93.0%, ARR 91.3%, versus just 3.7% of renewable credits. The tonnage-weighted mean composite score (0–100, higher = better) of redeemed credits (38.7) exceeded deposited credits (31.7) by 7 points, a 22% premium. As a result, BCT's net renewable share increased from 71% to 76% over the pool's lifetime.

The exit followed a type-level Gresham sequence (bad drives out good): the most valuable credit types were extracted first, in an order matching off-chain market demand ($\rho = -0.74$, $n = 7$ types; Supplementary Methods). Before the May 2022 crash, redemptions targeted high-value credits (tonnage-weighted quality 42.0); afterwards quality dropped to 32.4, reflecting that the premium credits had been extracted while their off-chain value still exceeded BCT's redemption cost.

Depositors and redeemers are almost entirely distinct populations: only 1.4% of 28,897 redeemer accounts also appear among the 509 depositor accounts. The top

10 redeemers account for 85.0% of extracted tonnage. Their transactions occurred in rapid bursts (median gap 4.6 seconds), consistent with programmatic execution.

This dual-margin structure is not predicted by standard adverse-selection models, which assume a single population: entry was indiscriminate and architecture-driven (99.6% tonnage pass-through), while extraction was deliberate and type-specialist.

Account forensics: who extracted and what they took

Account-level analysis of the 161 scored tokens' transfer histories reveals that the largest redeemers are type-specialist extractors, most of whom never participated in the deposit side.

Table 1 Top 5 redeemers by tonnage.

Account	Tonnes redeemed	Dominant type	Mean quality	Also depositor?
0x65a5...	651,334	Industrial gas	30.9	No
0x1b8e...	335,556	IFM	40.0	Yes
0xee4b...	195,051	REDD+	31.4	No
0x4b3e...	188,205	ARR	50.4	No
0x8556...	183,629	ARR	46.0	No

These five accounts removed 1.55 million tonnes: the vast majority of redeemed industrial gas, 37% of REDD+, and 44% of ARR. The largest, at 651,334 tonnes of industrial gas (42% of the total), is not an arbitrageur at all. It is a smart contract, an autonomous on-chain program, and it retired every credit it withdrew in the same transaction, permanently claiming them as offsets (verified on-chain; Data availability). That pattern is the signature of a retirement aggregator routing credits to final use. The other four are individual wallets. Their redeem-then-hold behavior is consistent with strategic arbitrage of the gap between BCT's uniform price and higher off-chain values: roughly \$7.4 million at midpoint price assumptions (\$10 million including the retirement contract; bounds \$0.3–\$22M; Methods). Fifteen of the twenty largest redeemers never deposited anything, and each targeted a single credit type.

Tracing each redeemed credit's immediate destination across the 20 largest extractors shows three pathways: cross-pool transfer to a quality-screened pool (NCT, Toucan's nature-restricted pool; 39.2%), immediate retirement (34.6%), and secondary-market sale (17.0%; Supplementary Methods).

Among the 399 accounts that both deposited and redeemed (a 1.4% overlap), a quality-swap analysis shows no systematic quality arbitrage (Supplementary Methods). The dual-margin mechanism thus operates through largely distinct account populations connected by the pool's uniform pricing. A first-funder audit of the top 20 accounts per margin found entity-level links for a minority: one shared funding source, direct credit transfers among three accounts, and two dual-margin accounts, together 26%/9% of top-20 deposit/redemption tonnage (Supplementary Table 5). The separation is therefore account-level, not entity-level.

Quality atlas and quality gating

BCT's PQD of 0.679 places it at the bottom of the VCM quality spectrum, which spans a 10-fold range from 0.076 (DACCS, direct air carbon capture and storage) to 0.759 (grid-connected renewables) (Fig. 3). A within-pool permutation test ($z = -0.64$, $p = 0.27$) finds no evidence of within-pool selection bias; the eligible universe was uniformly low-quality. Our letter grades B–AAA band the composite score (Methods); BBB separates credits with verified additionality from those without. Applied to the actual deposit stream, a BBB gate would have cut the deposit-weighted deficit from 0.689 to 0.506 while admitting only 7.2% of deposited tonnage (Fig. 4).

Temporal dynamics and robustness

Deposit quality declined over BCT's operating life (Spearman $\rho = -0.24$, $n = 1{,}187$). The decline is entirely a vintage effect: later deposits drew on progressively older vintages, the credit's emission-reduction year, not on intrinsically worse credits within a vintage (Supplementary Table 2). The May 2022 cryptocurrency crash provides exogenous deposit-side evidence of the causal channel: post-shock deposit quality fell 3.4 points (Cohen's $d = 0.49$, a moderate effect) as volume collapsed by 98% (Fig. 2b) and the quality-decline slope tripled, consistent with the price-to-quality feedback above (Supplementary Methods).

Pool design and asset fate: a within-token cross-pool contrast

A sharper contrast on the same data comes from 14 credit tokens (1.50 million tonnes) deposited into *both* the unscreened pool (BCT) and the quality-screened pool (NCT), so the same physical credit appears under both designs with quality, vintage, registry, and project type held fixed. At the pool level, these credits were fully redeemed from BCT but only 28.5% redeemed from NCT. Token by token (the unit of inference, $n = 14$), the mean redemption gap is +73.9 percentage points (95% CI [51, 93]; Fig. 5), unanimous in direction across every token and every credit type, with three exact tests (sign, permutation, Wilcoxon) returning $p = 2.4 \times 10^{-4}$. One selection effect qualifies the gap: the 14 are the redeemed members of the 31 shared credits, so the BCT-side 100% is partly built in while the NCT-side 28.5% is not; the +73.9-point figure is therefore an upper bound. This is a two-pool contrast, not 14 independent treatments; it shares the pool-level confounds that keep the two-pool analyses descriptive (Methods), and it covers only nature-based credits. We therefore read it as *strongly suggestive* of a pool-design effect rather than a proof, though robust: a hidden confounder would have to more than triple a credit's odds of redemption in one pool over the other to overturn the result (Rosenbaum $\Gamma = 3.05$; Supplementary Methods).

Cross-pool comparison. When we recover per-token credit identity on-chain for five pools across two operators (Toucan and C3, a second bridge protocol) and score each by our common framework, mean deposit quality rises monotonically with screening strictness, from the unscreened pools (∼29–31) through nature-screened pools (∼39–40) to the category-allowlist pool (77.9; Supplementary Table 4). This gradient is, however, definitional: a screen is defined on credit type and our composite is type-driven, so screened pools must score higher. A within-type check confirms the boundary: screened and unscreened pools hold nature-based credits of near-identical within-type quality (differences $\leq$0.32 points; Supplementary Methods), so screening acts on type, not within-type quality; the gradient is framework-derived (the C3 pools have no external validation). We therefore read the cross-pool pattern as corroborative and descriptive only, not as evidence that design causes quality.

Market-revealed quality hierarchy

Credit type alone, not our quality framework, captures the dominant axis of selective redemption. A type-only prediction rule out-predicts the quality-grade rule, and within the 116 renewable tokens neither vintage nor grade predicts redemption (Supplementary Table 3). The paper's core findings (composition, base-rate over-selection, redemption-side extraction, and the within-token comparison) accordingly rest on transaction data and Verra credit-type labels, not on the scoring framework (Supplementary Table 6).

The net result is that 9.3 of 9.6 million deposited B-grade tonnes remain unredeemed (the scored analysis covers 15.2 million tonnes), making BCT one of the largest carbon credit graveyards in existence.

Discussion

Here, transparency without price differentiation did not prevent quality collapse; we argue it cannot in general. Every transaction was public and every credit's quality metadata freely available on the Verra registry and the blockchain. Yet the market collapsed exactly as economic theory predicts for markets with *hidden* quality differences.

The underlying principle is Gresham's Law, which Akerlof generalized in 1970 as the "lemons problem" for markets where quality is unobservable[1]. BCT shows the same dynamic operates even when quality *is* observable: the mechanism we term design-enabled adverse selection runs through market architecture rather than information asymmetry, identifying a distinct pathway to lemons-like outcomes. Our within-token contrast provides credit-level descriptive evidence consistent with the lemons dynamics that Manshadi et al.[2] model theoretically.

Whenever heterogeneous assets are pooled at a single clearing price and the highest-quality units retain higher value outside the pool, those units are selectively withdrawn while low-quality units accumulate, *independently of whether quality is observable.* Observability changes who profits from the spread, not whether the spread exists. This incentive structure is well documented in traditional finance as cheapest-to-deliver pricing in the agency mortgage-backed-securities market[25,26,27,28]; BCT adds what that setting cannot show: full quality observability and a documented exit margin (Supplementary Discussion).

The misattribution of BCT's collapse to REDD+ credits[7,8,9] illustrates how structural quality failures can hide in plain sight. The CDM-era renewables that filled 69.1% of BCT had near-zero additionality: the projects were already economically competitive. Unlike REDD+ failures, which generate visible controversy over baselines (the no-project emission levels against which credits are measured) and deforestation, the additionality failure of renewable energy credits is structurally invisible: the dams and wind farms exist, the electricity is real. Quality assurance frameworks that focus on high-profile failure modes may miss the categories that dominate pool composition.

Over nine million tonnes of nominally valid offsets remain stranded to date (97.6% non-redemption within the scored subset). An illustrative welfare gap, the climate value lost relative to a counterfactual pool holding registry-average credits, has a Monte Carlo median of \$146M (90% simulation interval \$68M–\$252M, parameter uncertainty only). Meanwhile, a few type-specialist accounts captured an estimated \$10 million at midpoint assumptions (an upper bound; Results) by selectively redeeming the credits that off-chain markets still valued, though the single largest such address is a retirement contract, not a profit-taking arbitrageur. The two figures are complementary rather than overlapping: the welfare gap prices the pool's composition, while stranding describes what happened to its circulation. Paradoxically, BCT's failure functioned as an inadvertent quality filter, removing 9.3 million tonnes of low-quality credits from active circulation to date, an illiquidity event no registry-level intervention has matched.

This Gresham logic is not specific to carbon: it applies wherever heterogeneous assets are pooled at a uniform price[3,14]. The same lemons selection appears at the entry of six uniform-priced NFT vaults, pools of unique digital collectibles on NFTX, the largest such protocol. Deposited items sit below their collection's median value in all six vaults, strongly in five and marginally in the sixth (median percentiles 0.25–0.40, where 0.5 means no selection; each $p < 10^{-10}$). No vault shows extraction at exit. BCT, whose bridge passed heterogeneous credits wholesale, expressed selection at both margins (Supplementary Methods).

These findings apply most directly to voluntary pooling and registry-based bundling of credits, where quality-heterogeneous units clear at a common price[14,15]. The Article 6.4 mechanism is a registry-based crediting mechanism rather than a uniform-priced pool, so it inherits the design principle (price quality, or expect selection at any unpriced margin) rather than a direct prediction. Gating alone would not have sufficed: even a minimal quality floor (composite score $\geq$29) would have excluded 58% of BCT's tonnage. These results point to three design principles for pooled crediting mechanisms: quality-differentiated pricing tiers rather than single-price pools; gating on both margins, since exit extraction proved as damaging as entry loading; and quality floors that adjust as composition shifts. Our findings are consistent with, and lend urgency to, the ICVCM's Core Carbon Principles framework[16]. The emerging quality-differentiation ecosystem (Singapore's integration of independent ratings into eligibility assessments[17], blockchain credibility indices[18]) must incorporate dual-margin design to be effective; we release the three principles as an open-source reference implementation[19] (registry and gate on-chain, monitor off-chain; $\approx$19k gas; Supplementary Methods).

The same data also yield a deployable early warning. A cumulative Lemons Index (our running deposit-weighted PQD; 0–1, higher = worse), computed only from deposits already observed, crossed its 0.50 danger threshold at launch (0.71, price near its \$5.66 launch level), roughly seven months before the price fell below half the launch price. A framework-free variant, the cumulative renewable share, exceeded its threshold permanently from the pool's first week (Supplementary Methods). The signal is a standing level, not a trend: the diagnostic information was in the composition from inception. Paired with the quality gate above (0.689 to 0.506 at a BBB floor, 7.2% of tonnage admitted; Fig. 4), this converts the post-mortem into an actionable protocol: audit composition at inception, gate deposits and redemptions on quality, and re-audit as the pool evolves.

Several limitations warrant note. Our quality scores are author-derived (commercial ratings cover only a handful of BCT projects), but the core findings rely on transaction data and Verra credit-type classifications, not on the scoring framework. The framework itself is calibrated against CCP labels ($d = 1.8$; Fig. 6), commercial ratings ($\rho = +0.901$, $n = 27$; non-blind, scored with agency ratings in view; Supplementary Fig. 1), and a large-language-model panel (Fleiss' $\kappa = 0.6$, moderate agreement; Supplementary Methods). Both cross-pool comparisons are descriptive, not causal identifications. The within-token contrast (+73.9 pp gap, $\Gamma = 3.05$ bound) and the pool-level difference-in-differences (comparing changes across the two pools) rest on only two pools; their statistics quantify per-pair variability, not the pool-level confounds (exposure timing, liquidity, eligibility) that remain collinear with the screening label (Methods). We read the contrast as strongly suggestive of a pool-design effect, noting that the dominant renewables are structurally excluded from this nature-based comparison. One potential confound is, however, excluded: the bridge decomposition (99.6% pass-through by tonnage) shows that entry-side selection was architectural rather than strategic.

This result carries a falsifiable policy implication: any pooled crediting system providing quality transparency without quality-differentiated pricing will experience the same collapse. The transparency reforms by the ICVCM and the Article 6.4 Supervisory Body are necessary but not sufficient: if Article 6.4 pools heterogeneous credits at a single price, the same Gresham dynamic should emerge; in mechanisms with quality-differentiated tiers it should not. This prediction is testable as Article 6.4 issuance scales.

The tokenized real-world asset sector now exceeds $25 billion and is replicating pool-based architecture across commodities, bonds, and receivables. BCT is the first pool to complete a full quality-collapse cycle with auditable transaction data. Its lesson is straightforward: quality-differentiated design is not an optional refinement but a structural prerequisite for market integrity.

Methods

Full methods are provided in Supplementary Methods; this section summarizes the essentials.

Data. All 1,187 deposits to the BCT pool (October 2021 to late 2022; 168 Verra VCS projects, 345 TCO2 tokens, Toucan's per-project-vintage credit tokens; approximately 22 million tonnes) and all 708 NCT deposits were collected from Polygon (the blockchain hosting BCT) via direct node queries (RPC), recording transaction, depositor, token, project, vintage, and tonnage. Projects were cross-referenced against the Verra registry (methodology category, country, CCP status; eleven categories; tonnage-weighted statistics). Redemptions are ERC-20 Transfer events with the pool as sender (35,432 events across the 161 originally scored tokens). Daily BCT prices (1,493 observations, October 2021 to July 2026) come from the DeFi Llama API (a cross-exchange price aggregator) via the committed `fetch_bct_prices.py`.

Quality scoring. A composite score (0–100) weights six dimensions: removal type (0.250), additionality (0.200), MRV (monitoring, reporting, verification; 0.200), permanence (0.175), vintage (0.100), and registry/methodology (0.075), derived from the Oxford Offsetting Principles[20] and the ICVCM CCP Assessment Framework[16]; grades AAA–B are threshold bands, and seven disqualifiers cap grades. The Pool Quality Deficit is PQD $= 1 - \bar{C}/100$ (tonnage-weighted); its running deposit-weighted value is the Lemons Index used for early warning. Weight perturbation (10,000 random reweightings summing to one) leaves 93.7% of grades unchanged.

Selection tests. The base-rate selection coefficient $S = f_{\mathrm{BCT}}/f_{\mathrm{VCS}}$ (central base rate 37%, sensitivity range 26–48%) is tested with account-clustered inference. The within-token contrast treats the 14 tokens deposited into both BCT and NCT as their own controls (three exact tests plus a percentile bootstrap); the estimand is the pool-design effect on dual-eligible credits, with a Rosenbaum-style Γ for unobserved confounding. A deposit-level difference-in-differences and a five-pool, two-operator comparison are reported as descriptive corroboration only.

Price, cross-domain, and welfare. A first-differenced OLS with Newey–West (autocorrelation-robust) errors tests price-quality co-movement ($n = 330$ daily changes); weekly Granger tests ($n = 55$) are exploratory. The NFTX cross-asset pipeline assigns each minted (deposited) or redeemed item the percentile of its nearest sale within a drift-controlled window of collection sales. Entry-margin selection is tested per vault (Wilcoxon) and across vaults (sign test), exit-margin extraction by Mann–Whitney rank tests; router (intermediary batching) accounts are excluded. The welfare gap is a 10,000-iteration Monte Carlo simulation (chosen to propagate several uncertain inputs jointly): literature additionality ranges[4,12,22,23], the social cost of carbon (\$50–\$200/tCO_2), and a base-rate counterfactual composition.

Inference. All p-values were corrected using the Benjamini–Hochberg false-discovery-rate (FDR) procedure at $\alpha = 0.05$ across ten primary tests: (1) base-rate selection coefficient (account-clustered; naive binomial as reference only), (2) full-sample and (3) pre-shock temporal correlations, (4) the within-token sign test, (5) the pool-level difference (descriptive), (6–7) bidirectional Granger, (8) within-pool permutation, (9) depositor concentration, and (10) the redemption differential (token-level Mann–Whitney; $n = 161$). A cluster-robust bootstrap (by account, 10,000 iterations) covers deposit-level statistics.

Data availability

All data, scoring rubrics, analysis scripts, and smart contract source code are available at https://github.com/Adeline117/carbon-neutrality (version tag `v1.0-erl-submission` pins the state cited here) under an MIT license. Machine-readable rubrics are in JSON format. The 318-credit methodology batch dataset with per-dimension scores is included. All 345/345 BCT tokens are scored (161 original + 184 imputed). On-chain deposit data for both BCT (1,187 transactions) and NCT (708 transactions) are provided with transaction hashes enabling independent verification. The externally-owned-account versus smart-contract status of the top redeemers was checked with `eth_getCode` against a public Polygon node; results are recorded in `redeemer_contract_check.json`. The entity-level independence audit (first-funder resolution for the top 20 accounts per margin) is reproducible via `entity_funding_analysis.py`, with per-wallet results in `entity_funding_analysis.json`. LLM panel outputs for all models across 29 credits are included. Analysis scripts are pure Python with NumPy/SciPy as sole dependencies; a single verification script (`tools/verify_headline_numbers.py`) re-asserts every headline number against the cached analysis outputs.

Code availability

All analysis code is available at https://github.com/Adeline117/carbon-neutrality under an MIT license.

Author contributions

A.W. designed the study, collected and analyzed all data, developed the quality scoring framework, performed all statistical analyses, and wrote the manuscript. W.C. supervised the research and provided critical revisions.

Competing interests

The authors declare no competing interests.

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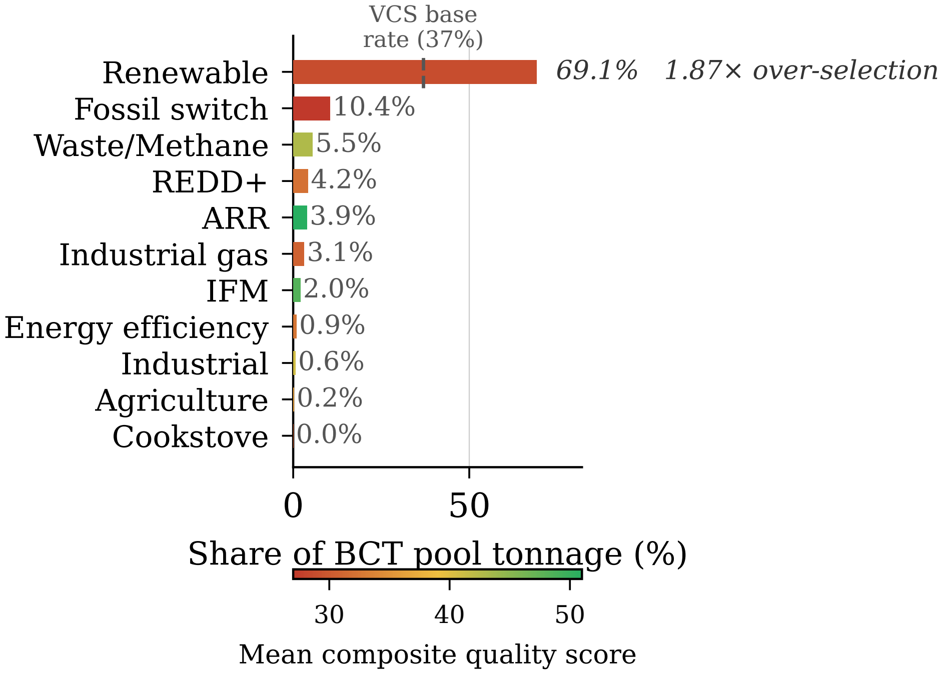

Fig. 1 BCT pool composition by methodology category, weighted by deposited tonnage. Renewable energy credits constitute 69.1% of total deposited tonnage; REDD+ constitutes just 4.2%.

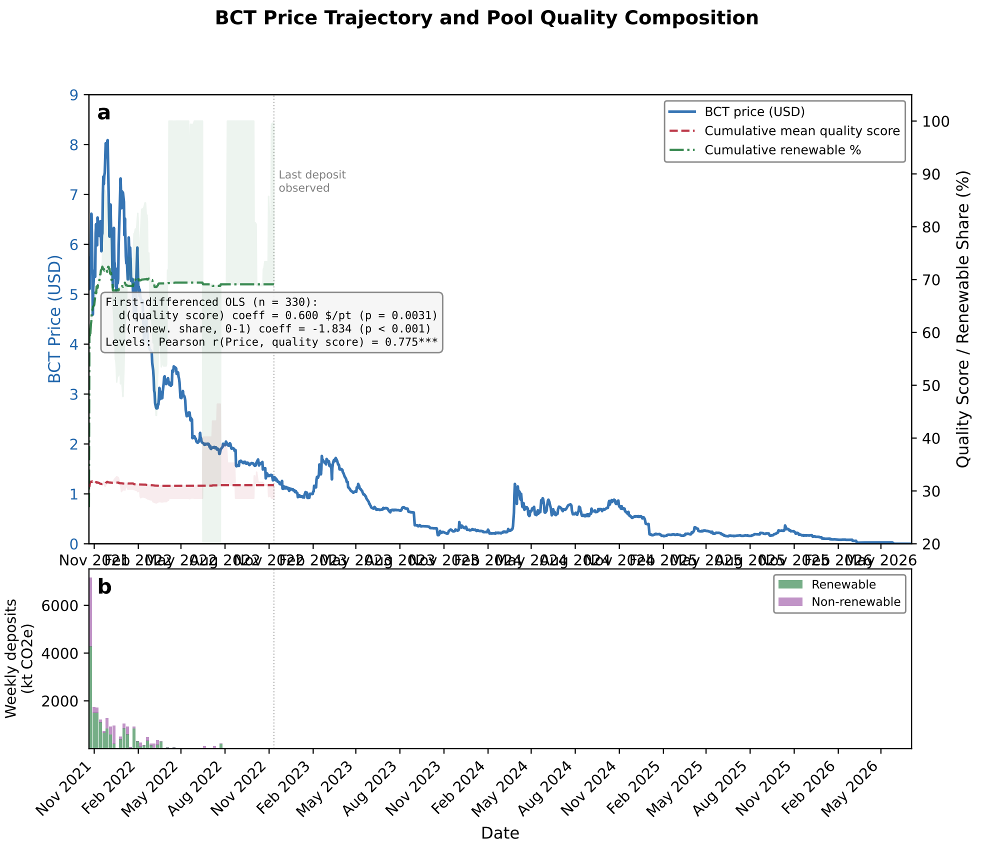

Fig. 2 BCT price trajectory and pool quality composition. (**a**) Daily BCT price against cumulative Pool Quality Deficit and cumulative renewable share; inset reports the first-differenced OLS (n = 330) and the levels correlation (Pearson r = 0.774). (**b**) Weekly deposit volume split into renewable and non-renewable tonnage.

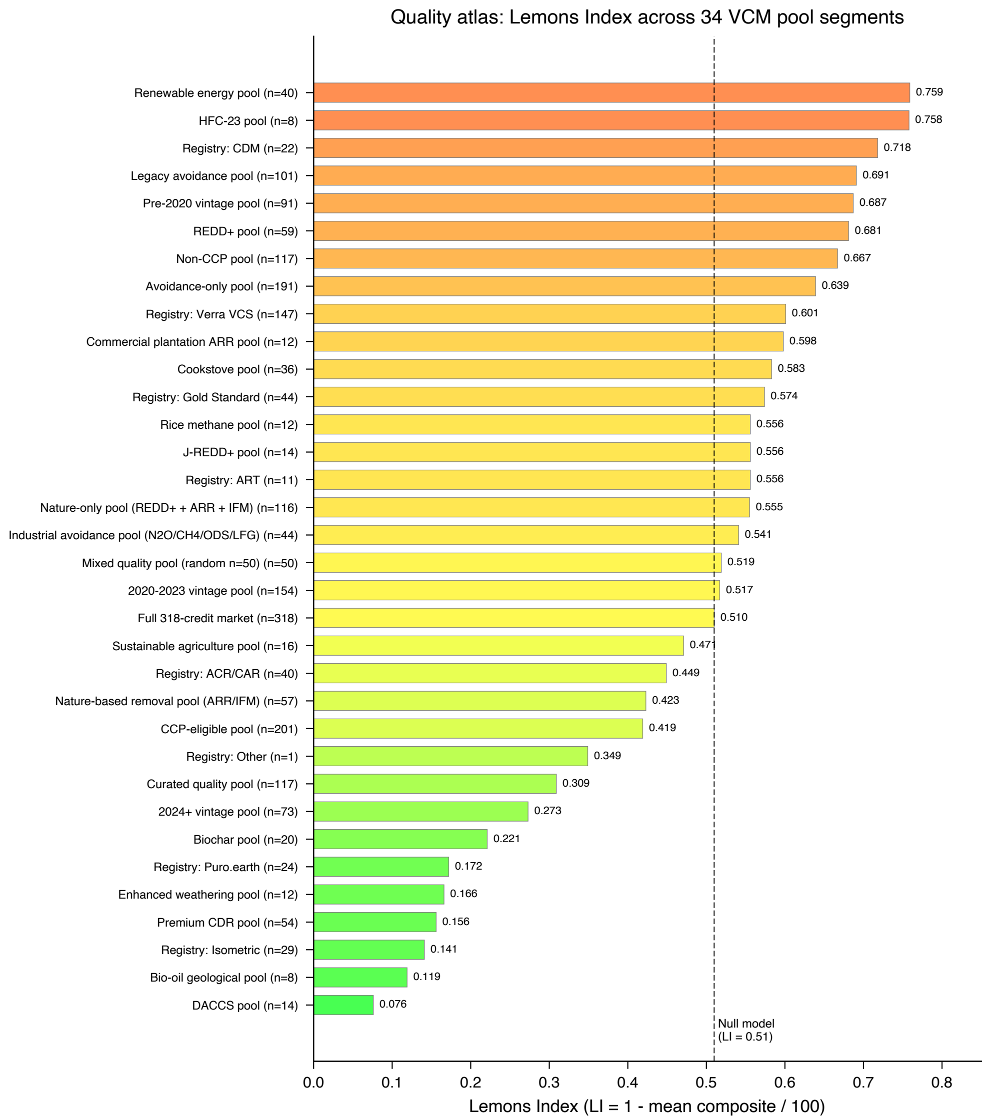

Fig. 3 Quality atlas: 34-segment PQD spectrum with null model baseline. PQD ranges from 0.076 (DACCS) to 0.759 (grid-connected renewables).

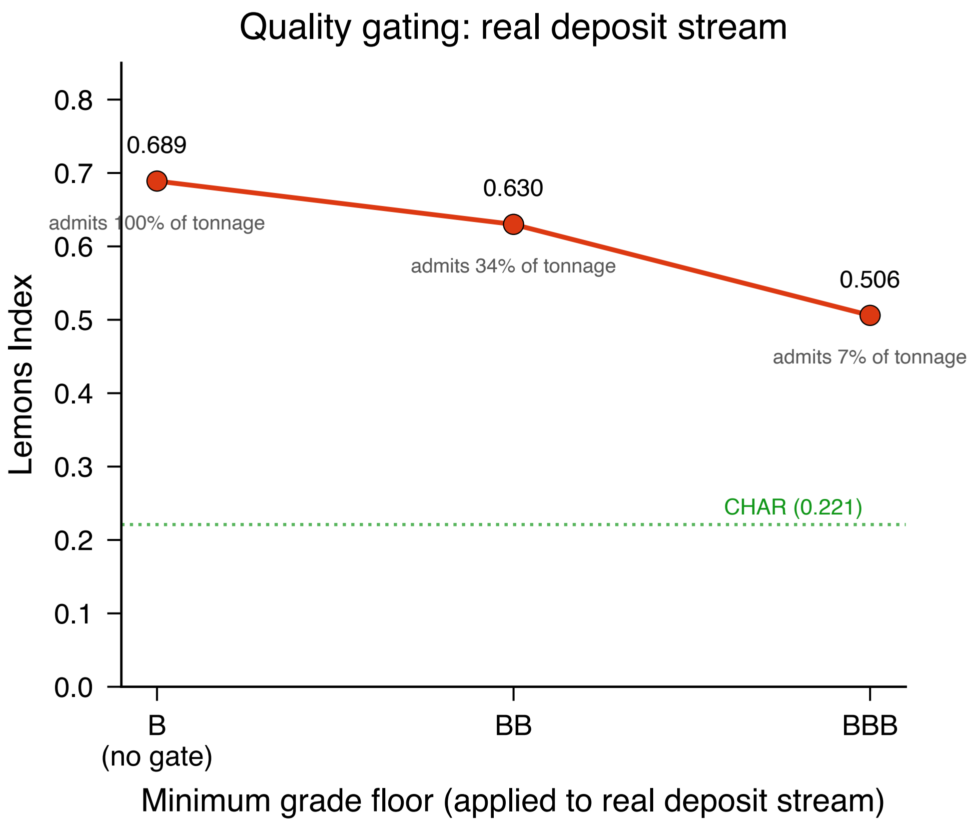

Fig. 4 Quality gating applied to the real BCT deposit stream: gated Lemons Index and admitted tonnage share under grade floors (B = no gate, BB, BBB). A BBB floor cuts the deposit-weighted Lemons Index (PQD) from 0.689 to 0.506 but admits only 7.2% of deposited tonnage; the CHAR category-allowlist pool (0.221) is shown for reference.

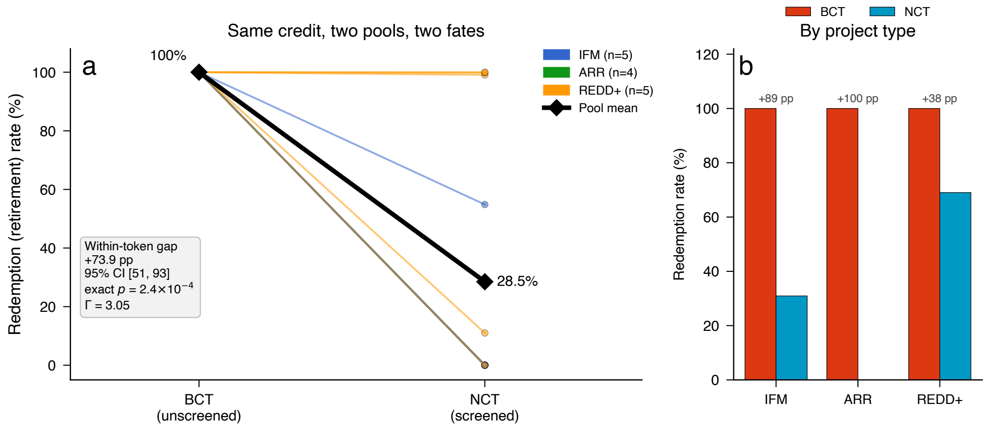

Fig. 5 Within-token cross-pool contrast: redemption of identical credits under two pool designs (descriptive; a two-pool comparison, not a causal identification). (**a**) Per-token slope plot of redemption (retirement) rate for the 14 carbon credit tokens deposited into *both* the unscreened BCT pool and the quality-screened NCT pool, colored by project type; the bold black line traces the pool means (100.0% redeemed from the unscreened pool versus 28.5% from the screened pool). Across the 14 pairs (the unit of inference) the mean within-token redemption gap is +73.9 percentage points (token-level bootstrap 95% CI [51, 93]); the direction is unanimous, so exact sign, permutation, and Wilcoxon tests all return $p = 2.4 \times 10^{-4}$ whether or not the one near-tied credit is included. (**b**) Redemption rate by project type (bars: tonnage-weighted rates; annotated gaps: per-token mean differences): the gap is positive within every type (IFM +89, ARR +100, REDD+ +38 pp per-token). The result is robust to unobserved confounding up to a Rosenbaum bound of $\Gamma = 3.05$. Depositor-level temporal, base-rate, and difference-in-differences robustness analyses are reported in Supplementary Methods.

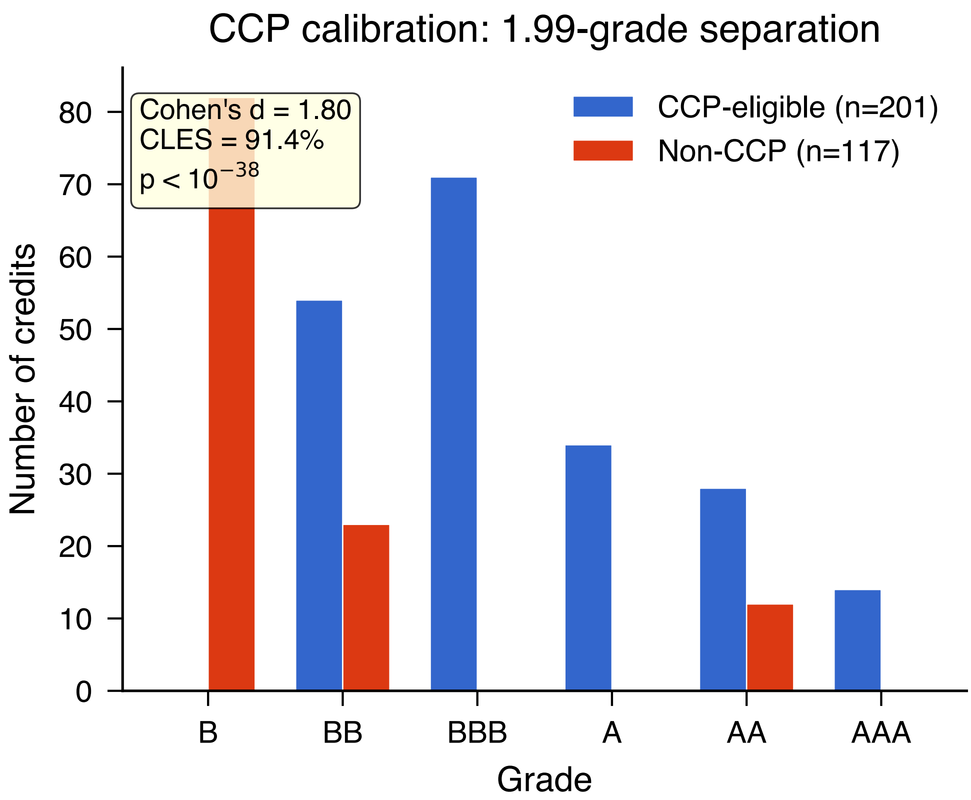

Fig. 6 CCP calibration: quality score distributions for CCP-eligible versus non-CCP credits. Cohen's $d = 1.8$ (95% CI [1.5, 2.16]). In-figure statistics: the title's 1.99 grades is the mean grade-band separation between the two groups; CLES is the common-language effect size (probability a random CCP-eligible credit outscores a random non-CCP credit, 91.4%).